

■ BACKUP SYSTEM SETUP GUIDE

Automated Backup Plan using Windows Task Scheduler & Google Drive Sync
Nibras International School — IT Department Portfolio Project

About This Guide

This step-by-step guide demonstrates two backup methods for school data: **Method A** uses Windows Task Scheduler to automate local/network backups, and **Method B** uses Google Drive for Sync cloud backup. Together they form a 3-2-1 backup strategy: 3 copies of data, on 2 different media, with 1 offsite (cloud). This guide was prepared as a portfolio project by the IT Support staff at Nibras International School.

■ WHY BACKUP MATTERS FOR SCHOOLS

Risk	Scenario	Severity
Hardware Failure	Hard drive crash destroys all student results, attendance & exam papers permanently	CRITICAL
Accidental Deletion	Staff accidentally deletes question papers 1 day before exams	HIGH
Ransomware/Virus	Malware encrypts all school files and demands payment	CRITICAL
Power Surge	Sudden power cut corrupts open database files	MEDIUM
Theft	School laptop with student data stolen during holidays	HIGH
Fire/Flood	Physical disaster destroys computers and local drives	CRITICAL

Solution: The 3-2-1 Backup Rule

	3 copies of every important file (original + 2 backups)
	2 different storage types (e.g., local drive + external drive)
	1 offsite/cloud backup (Google Drive, so safe even if school burns)

■ METHOD A: Automated Local Backup with Windows Task Scheduler

This method creates a scheduled batch script that automatically copies important school folders to an external/network drive every night at 11:00 PM.

STEP 1 — Create the Backup Batch Script

Open Notepad and paste the following script. This script copies school data to a backup drive:

```
@echo off
:: Nibras International School – Auto Backup Script
:: Created by: IT Support Staff
SET SOURCE=C:\NibrasIS_SchoolData
SET DEST=D:\Backups\NibrasIS
SET DATE_STR=%date:~10,4%-~4,2%-~7,2%
SET DEST_FOLDER=%DEST%\%DATE_STR%
echo Starting backup: %DATE_STR%
echo Source: %SOURCE%
echo Destination: %DEST_FOLDER%
:: Create destination folder with today's date
mkdir "%DEST_FOLDER%" 2>nul
:: Copy all important school folders
xcopy "%SOURCE%\01_Students" "%DEST_FOLDER%\01_Students\" /E /H /Y /Q
xcopy "%SOURCE%\02_Academic" "%DEST_FOLDER%\02_Academic\" /E /H /Y /Q
xcopy "%SOURCE%\03_Administration" "%DEST_FOLDER%\03_Administration\" /E /H /Y /Q
xcopy "%SOURCE%\04_Staff" "%DEST_FOLDER%\04_Staff\" /E /H /Y /Q
:: Delete backups older than 30 days to save space
forfiles /p "%DEST%" /d -30 /c "cmd /c rmdir /s /q @path" 2>nul
echo Backup completed successfully: %DATE_STR%
echo Log saved to: %DEST%\backup_log.txt
>> "%DEST%\backup_log.txt" echo Backup: %DATE_STR% %TIME% - SUCCESS
```

Save as: NibrasIS_Backup.bat (on Desktop or C:\Scripts\)

STEP 2 — Schedule the Script in Windows Task Scheduler

2a	Open Task Scheduler	Press Windows + R → type taskschd.msc → press Enter
2b	Create Basic Task	In right panel, click 'Create Basic Task...' → Name: NibrasIS Nightly Backup → click Next
2c	Set Trigger	Choose Daily → Set Start Time: 11:00 PM → click Next
2d	Set Action	Choose 'Start a program' → Browse to NibrasIS_Backup.bat → click Next
2e	Finish & Test	Click Finish → Right-click task → 'Run' to test it manually → Check D:\Backups\NibrasIS for new folder
2f	Verify Log	Open D:\Backups\backup_log.txt → Confirm 'SUCCESS' entry appears after each run

STEP 3 — Recommended Backup Schedule

When	What to Back Up	Method
Nightly (11 PM)	All student results, exam papers, notices, admin records	Task Scheduler (auto)
Weekly (Friday)	Full school data including staff folders & IT assets	Task Scheduler (auto)
Monthly (1st)	Full archive backup to external USB drive stored in principal's office	Manual (IT staff)
Before Exams	Manual backup of all question papers to 2 separate drives + email to principal	Manual (IT staff)
End of Term	Archive entire term data to _ARCHIVE folder before new term begins	Manual (IT staff)

■ METHOD B: Cloud Backup with Google Drive for Desktop

Google Drive for Desktop (formerly Backup & Sync) keeps school files automatically synced to the cloud. This provides the offsite backup in our 3-2-1 strategy. Files deleted by mistake can be recovered from cloud Trash for up to 30 days.

Step 1	Download & Install	Visit drive.google.com/drive/downloads → Download 'Google Drive for Desktop' → Install on school computer
Step 2	Sign In with School Account	Use the school Google Workspace account (e.g., it@nibrasinternational.edu.bd) — NOT a personal Gmail account
Step 3	Choose Folders to Sync	In Drive settings → 'My Computer' tab → 'Add Folder' → Select C:\NibrasIS_SchoolData → Enable 'Sync with Google Drive'
Step 4	Set Sync Preferences	Choose 'Mirror files' mode so files are accessible both locally and on cloud. Set bandwidth limit if school internet is slow.
Step 5	Verify Sync Status	Look for Google Drive icon in taskbar. Green checkmark = synced. Click it to view recent uploads and confirm all folders appear.
Step 6	Test Recovery	Delete a test file from PC → Go to drive.google.com → Check Trash → Confirm file is there → Restore it to verify recovery works.
Step 7	Share with Principal	In Google Drive, right-click the school folder → Share → Add principal's email → Viewer access only (for oversight, not editing)

■ METHOD A vs METHOD B — Quick Comparison

Feature	Method A (Task Scheduler)	Method B (Google Drive)
Setup Complexity	Medium — requires batch script	Easy — GUI-based installation
Cost	Free (uses existing hardware)	Free (15 GB) or paid Workspace plan
Internet Required	No — works offline	Yes — needs internet to sync
Storage Location	Local drive / network drive	Google's cloud servers
Protection from Fire/Theft	■ No — same location	■ Yes — offsite cloud
Recovery Speed	Fast — local copy	Depends on internet speed
Automatic Scheduling	■ Yes — Task Scheduler	■ Yes — continuous sync
Recommended For	Nightly data backup, exam archives	Daily working files, easy access

Recommendation: Use BOTH methods together. Method A for nightly structured backups of critical data (results, exam papers, staff records). Method B for continuous cloud sync of daily working files. This gives you the complete 3-2-1 backup protection.

■ Monthly Backup Verification Checklist

IT staff should complete this checklist on the 1st of each month and file it with the principal:

■	Check D:\Backups folder — confirm last 30 days of backup folders exist	Date: _____
■	Open backup_log.txt — verify all entries show SUCCESS (no FAILED entries)	Verified by: _____
■	Test file restoration — copy 1 file from backup to desktop, open & confirm it works	Test file: _____
■	Check Google Drive — confirm school folder shows today's sync date & green checkmark	GB used: _____
■	Check Google Drive storage — if >80% full, notify admin to upgrade plan	Status: _____
■	Copy monthly archive to USB drive — label: 'NibrasIS_Archive_YYYY-MM' — store with principal	USB Label: _____
■	Confirm confidential folders (Finance, Staff Contracts) are NOT synced to personal accounts	Confirmed: _____
■	Update Equipment Log with any hardware changes affecting storage capacity	Changes: _____

Checked by (IT Staff): _____

Date: _____

Signed by (Principal): _____

"Indeed, Allah loves those who are constantly repentant and loves those who purify themselves." (Quran 2:222)

Just as we purify ourselves spiritually, we protect and preserve the trust (Amanah) placed in us — including the school's digital records. Proper backup is not just good IT practice; it is fulfilling our responsibility to the school, students, and community of Nibras International School.